

Downstream analysis: age structure and combinatorial risk factors

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Outline

1. Why cancer epidemiology and age structure is interesting (3 slides)
2. How to study both genomic alterations and age structure simultaneously (4 slides)
3. What data I actually need (2 slides)

Age structure and biallelic loss



How many *clonal, rate-limiting* mutations are there?

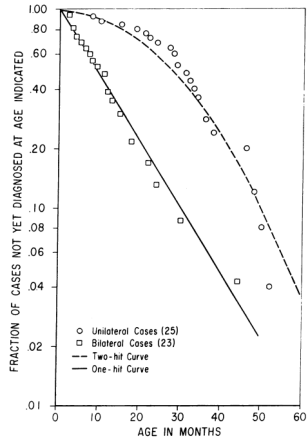
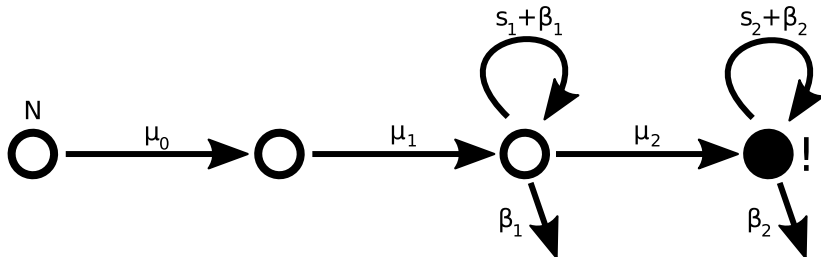


FIG. 1. Semilogarithmic plot of fraction of cases of retinoblastoma not yet diagnosed (S) vs. age in months (t). The one-hit curve was calculated from $\log S = -t/30$, the two-hit curve from $\log S = -4 \times 10^{-5} t^2$.

¹AG. Knudson, PNAS 68.4 (1971): 820-823.

²F. Michor, Y. Iwasa and MA. Nowak, Nature Reviews Cancer 2004; 4: 197-205 doi:10.1038/nrc1295

Classic age structured models



- ▶ The survival curve $S(a)$ determines the incidence curve $I(a)$
- ▶ To fit (or “train”) the model to longitudinal data, we use a model for $S(a)$
- ▶ Model parameters: mutation rates μ_k , “fitnesses” s_k , and cell turnover/renewal rates β_k

But...

Limitations of age structured models

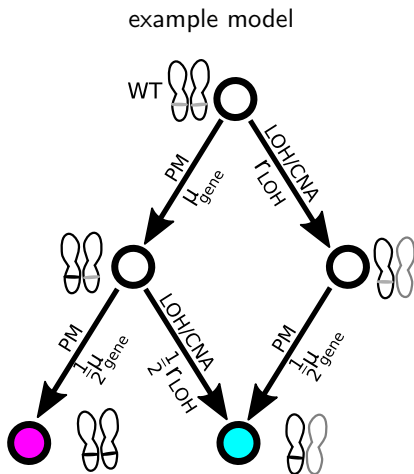
Lots of biologically relevant stuff we have learned since doesn't fit into these models:

1. can't distinguish between e.g. structural variants and SNVs
2. can't distinguish between different orders of events.

different combinations of alterations (e.g. loss-of-function indel and copy number loss) can result in biallelic loss, but may also represent different degrees of pathogenicity. IN BRIEF: *genomic* risk factors

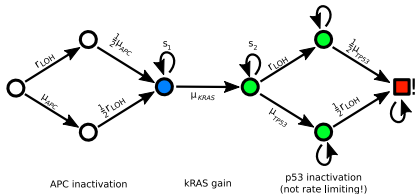
Generalised: Age structured models on graphs

1. At **specific loci**, different mechanisms of interest (SNVs, LOH, CNA, etc.) can happen in different orders. The different combinations of events can be represented with a graph.
2. We need to evaluate $S(a)$ for a model defined on a graph (right)
3. We have a new algorithm for doing this and getting MLE estimates of the parameters



This can model the age-incidence of arbitrarily granular subtypes. Whatever callers we can run and classify patients, we can get an age-incidence curve.

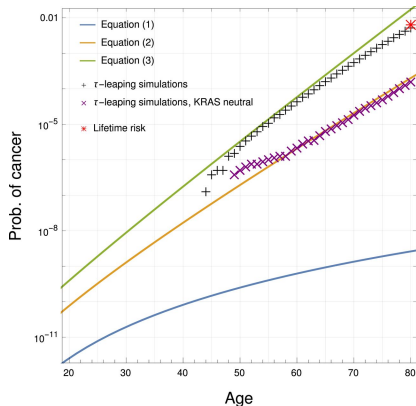
Colorectal adenocarcinoma model



- ▶ Can use A-D type approximation, or stochastic simulations

but:

- ▶ Mean-field breaks down at old ages / large probabilities
- ▶ Stochastic simulations are *extremely slow*



Kolmogorov forward equations

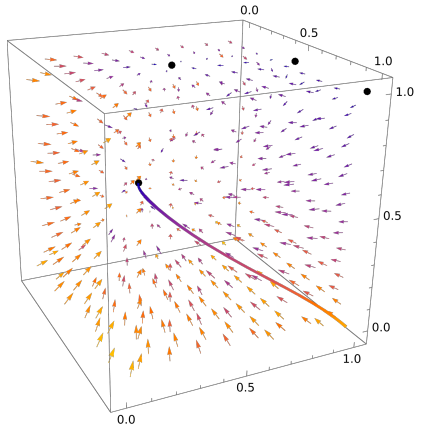
Using the big generating function Ψ , find the corresponding wave equation:

$$\frac{d\Psi}{dt} = \mathcal{H}\Psi \quad (1)$$

This can be solved using the method of characteristics, so we can instead integrate

$$\frac{d\vec{\gamma}}{dt} = \vec{X} \quad (2)$$

numerically, using an appropriate time stepper.

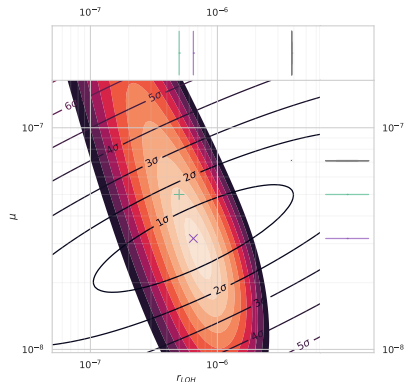
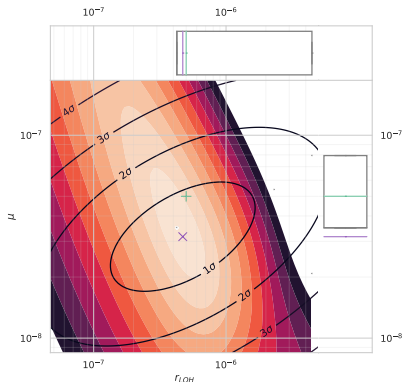


The vector field $\vec{X}$ and a characteristic $\vec{\gamma}$

Fast forward method

Parameter inference

Efficiently computing $S_k(a)$ means we can evaluate likelihood functions directly, just sampling them.....



What data I actually need

Basically, I want to structure on specific alterations. Here's a proposal to get us started:

- ▶ Focus on p53 (e.g.!) and for each patient with biallelic loss:
 1. establish the true WT for each patient as far as possible!
 2. call copy number changes (Battenberg), (non-silent) SNVs (Caveman?), and SVs (BRASS?).
 3. consensus?
- ▶ Thank you Ginny for the SOP spreadsheet
- ▶ Battenberg is also partly done (but be careful: back to Marian)
- ▶ Genomic signatures are also interesting and not locus-specific: I'm especially interested in seeing SBS1 called as it's clock-like.

What data I actually need

End result should be (for each locus of interest) a “spreadsheet” that looks like:

Patient ID	Age at diag.	17p major CN	17p minor CN	SV (over p53)	SNV (on p53)	indels...	.
B120002	68	2	0	None	...TCG>TAG...	...	.
B120005	54	2	1	None	None	such&such	.
...	...	...	...	...	...	...	.

NB: This “row” can be continued for any other loci/gene of interest.

Each column heading (except ID and age) becomes an edge on the graph and each combination of events becomes a final node on the graph. (Continuing the table corresponds to taking a Kronecker product of multiple graphs)